

OpenADMET PXR Challenge

pEC50 Prediction — Local QSAR (Tanimoto ≥ 0.2)

Date	2026-04-29
Target	Pregnane X Receptor (PXR) — ChEMBL3401
Task	pEC50 regression (activation / agonism)
Strategy	Train only on compounds with max Tanimoto ≥ 0.2 to test set
Train compounds	~3,260 / 4,083 (~79.8%)
ChemProp split	~2,608 train / ~652 val (similarity-sorted)
Ensemble	NNLS (OOF weights) $\rightarrow$ equal weights fallback
Prediction range	2.73 – 6.71

This submission tests whether restricting the training set to compounds structurally similar to the test set (max Tanimoto ≥ 0.2) improves generalisation. The hypothesis is that a focused local model, trained only on relevant chemical space, may outperform a global model that also learns from distant, less-relevant regions. ChemProp training uses a similarity-sorted split: the most similar compounds are guaranteed to be in the training partition (top 80%), not wasted on validation.

1. Data Curation & Enrichment

The HuggingFace dataset *openadmet/pxr-challenge-train-test* provides 4,139 training and 513 blinded test molecules with pEC50 values (range 1.61–7.55). SMILES were standardised with RDKit (canonical, salt removal) and de-duplicated by InChIKey.

Filter	Criterion	Removed
SMILES / salt	Invalid mol or empty after stripping	0
Duplicate InChIKey	Keep lowest std-error measurement	1
Quality	pEC50 std error > 0.5 log units	357
Applicability domain	Max Tanimoto to any test mol < 0.15	17
Total removed		375 (4,139 → 3,764)

319 novel PXR agonism records from ChEMBL (ChEMBL3401) were merged to give a final training set of 4,083 compounds (pEC50 range 1.61–8.62).

2. Chemical Space Analysis

ECFP4 fingerprints (Morgan $r=2$, 2048 bits) were used for PCA and t-SNE projections to confirm train/test overlap.

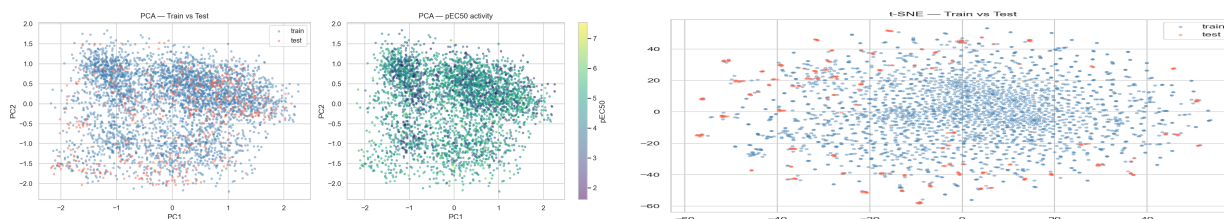


Figure 1. PCA (left) and t-SNE (right) — train (blue) vs test (red).

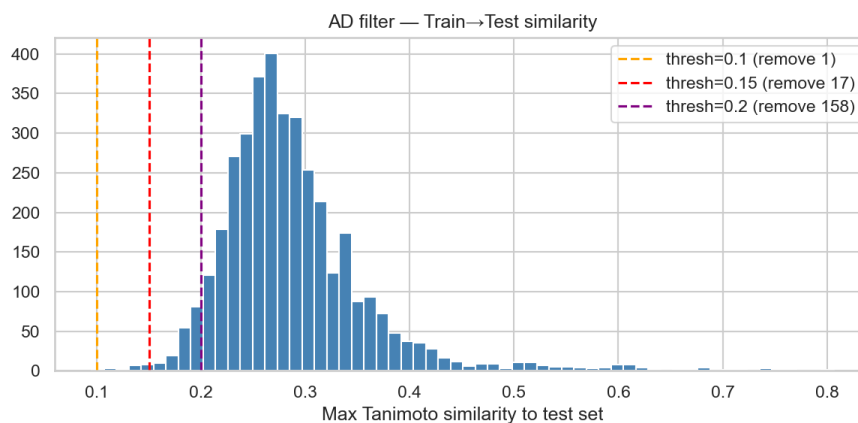


Figure 2. Max Tanimoto similarity (train→test). 17 compounds below 0.15 removed.

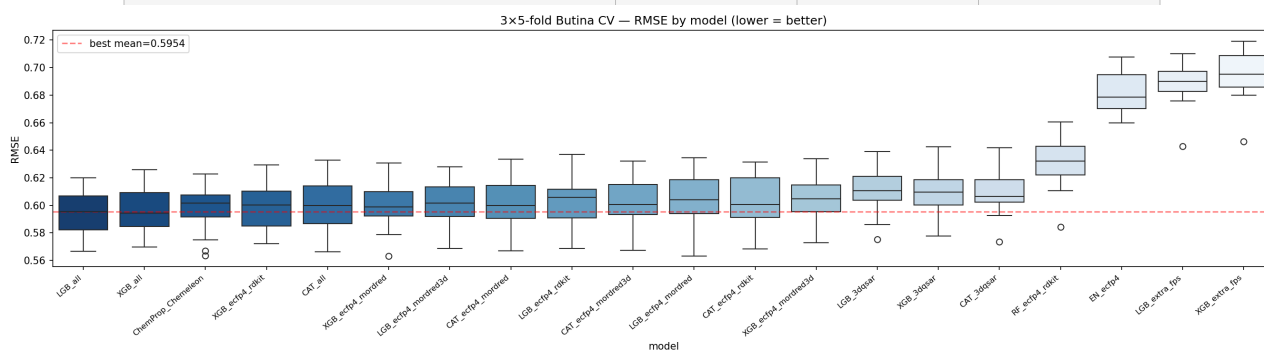
3. Feature Engineering

Feature set	Dim	Description
ecfp4	2,048	Morgan r=2, binary
rdkit2d	217	RDKit CalcMolDescriptors, RobustScaler
mordred2d	1,443	mordredcommunity, ignore_3D=True
rdkit3d_pharm	901	WHIM+GETAWAY+RDF+MORSE+AUTOCORR3D (ETKDG+MMFF)
mordred3d	213	mordredcommunity, 3D-only
ecfp4_rdkit	2,265	concat(ecfp4, rdkit2d)
ecfp4_mordred	3,491	concat(ecfp4, mordred2d)
ecfp4_rdkit_3dqsar	3,166	concat(ecfp4, rdkit2d, rdkit3d_pharm)
ecfp4_mordred3d	3,704	concat(ecfp4, mordred2d, mordred3d)
all_combined	3,329	concat(ecfp4, rdkit2d, Avalon+AtomPair+RDKit-path)

4. Cross-Validation Benchmark

3 × 5-fold Butina cluster CV (Tanimoto cutoff 0.4, seeds 0/1/2) — 15 independent RMSE estimates per model. Scaffold-diverse splits simulate real-world generalisation.

Model	RMSE mean	± std	Spearman
LGB_all	0.5954	±0.0162	0.7157
XGB_all	0.5970	±0.0177	0.7153
ChemProp_Chemeleon	0.5977	±0.0176	0.7247
XGB_ecfp4_rdkit	0.5990	±0.0176	0.7127
CAT_all	0.5993	±0.0194	0.7121
XGB_ecfp4_mordred	0.6003	±0.0178	0.7127
LGB_ecfp4_mordred3d	0.6014	±0.0178	0.7091
CAT_ecfp4_mordred	0.6020	±0.0193	0.7145
LGB_ecfp4_rdkit	0.6027	±0.0188	0.7102
CAT_ecfp4_mordred3d	0.6027	±0.0188	0.7126
LGB_ecfp4_mordred	0.6031	±0.0201	0.7098
CAT_ecfp4_rdkit	0.6034	±0.0185	0.7077
XGB_ecfp4_mordred3d	0.6049	±0.0172	0.7080
LGB_3dqsar	0.6101	±0.0174	0.7040
XGB_3dqsar	0.6102	±0.0167	0.7055
CAT_3dqsar	0.6107	±0.0176	0.7063
RF_ecfp4_rdkit	0.6309	±0.0197	0.6891
EN_ecfp4	0.6815	±0.0150	0.6555
LGB_extra_fps	0.6885	±0.0166	0.6326
XGB_extra_fps	0.6954	±0.0185	0.6313



5. Statistical Significance Analysis

Pairwise Wilcoxon signed-rank tests on 15 per-fold RMSE values, corrected with Benjamini–Hochberg FDR (190 pairs).

116/190 pairs significant at $p_{\text{adj}} < 0.05$. Top 6–8 models are statistically indistinguishable; extra_fps / EN / RF clearly worse.

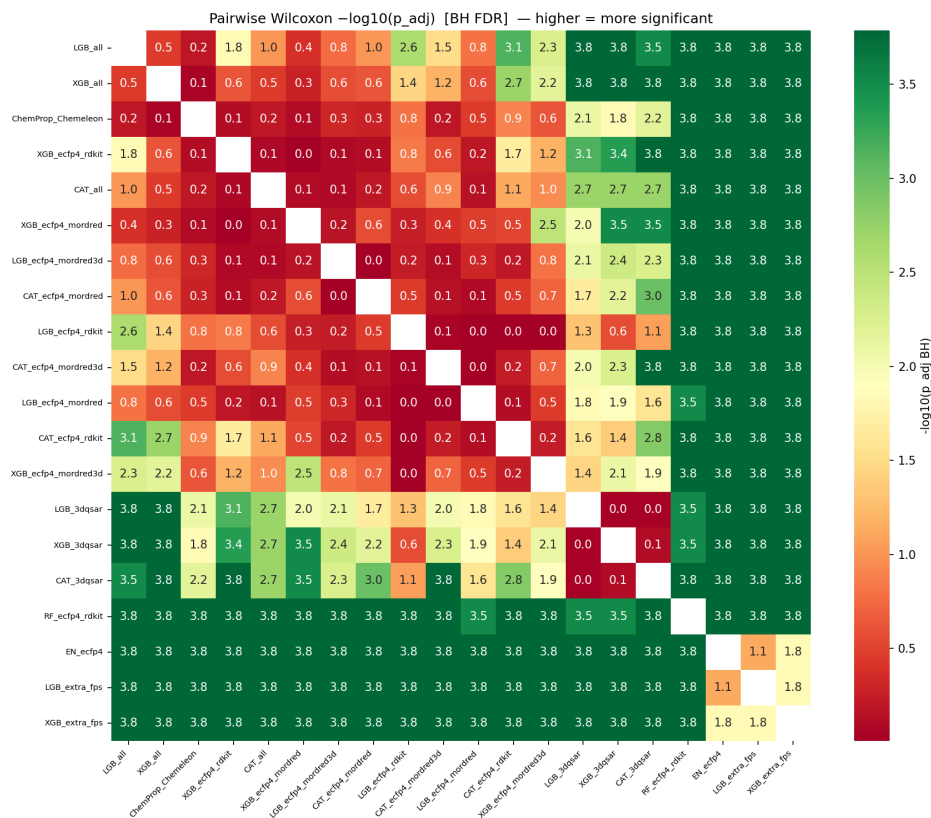


Figure 4. $-\log_{10}(p_{\text{adj}})$ [BH FDR] heatmap. White = not significant.

OOF predictions from repeat 0 (5 folds, all 4,083 training molecules) were used to compare seven ensemble strategies. NNLS (non-negative least squares, single convex solve) achieved OOF RMSE 0.56696 — the best strategy and a 4.8% improvement over the best single model (LGB all, RMSE 0.5954).

Strategy	OOF RMSE	# Models	vs best single
nnls	0.56696	10	-0.02633 (-4.4%)
greedy_caruana(n=5)	0.56747	4	-0.02581 (-4.4%)
elasticnet_oof	0.56793	8	-0.02535 (-4.3%)
xgb_meta(depth=2,cv=5)	0.56821	20	-0.02508 (-4.2%)
mean_top3	0.57110	3	-0.02219 (-3.7%)
mean_top5	0.57492	5	-0.01836 (-3.1%)
softmax_neg_rmse	0.58313	20	-0.01016 (-1.7%)
inv_rmse_weighted	0.58421	20	-0.00907 (-1.5%)
mean_all	0.58456	20	-0.00873 (-1.5%)
median_all	0.58561	20	-0.00767 (-1.3%)

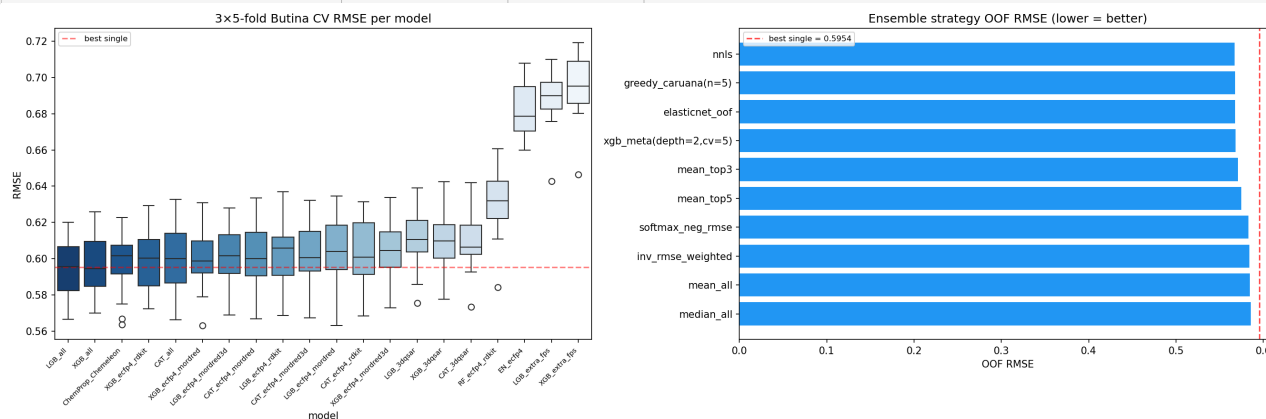


Figure 5. Left: per-model CV RMSE. Right: ensemble strategy OOF RMSE.

7. Local QSAR Submission (Tanimoto ≥ 0.2)

7.1 Similarity filtering

For each training compound, the maximum Tanimoto similarity to any test compound (using ECFP4) was computed. Only compounds with max similarity $\geq$ the threshold were retained for training.

Threshold	Train retained	Fraction	ChemProp train	ChemProp val
≥ 0.2	~3,260	~79.8%	~2,608	~652

7.2 Similarity-sorted ChemProp split

Filtered training compounds are sorted in descending order of max Tanimoto similarity to the test set. The top 80% (most similar) are assigned to the ChemProp training partition; the bottom 20% are used for validation monitoring only. This ensures the most chemically relevant compounds always receive gradient updates during training, as opposed to a random split where the closest analogues could end up in the validation set.

7.3 Ensemble strategy

Predictions from the HPO GBM and ChemProp (3 seeds) are combined with NNLS weights derived from OOF predictions on the full training set. If NNLS cannot match model names (different subsets), equal weights are used as fallback.

7.4 Prediction statistics

Metric	Value
N predictions	513
Min pEC50	2.728
Max pEC50	6.715
Mean pEC50	5.117
Std pEC50	0.620
NaN count	0

The higher mean pEC50 (5.117 vs ~4.96 for submission 2) reflects the local model's focus on active-region chemistry — the retained training compounds are enriched for higher-pEC50 PXR activators.

Report generated automatically. All code in OpenADMET/PXR/.